

1. ALGEBRAIC EXPRESSIONS 1

This chapter lays the groundwork for manipulating algebraic expressions. You will learn to work with powers (indices), expand brackets, factorise expressions, handle negative and fractional exponents, manipulate surds, and rationalise denominators. Every technique is interrelated, and mastering these skills is essential for higher mathematics.

Learning Objectives

- **Multiply and divide integer powers (pages 2–4):** Understand and apply the laws of indices in multiplication and division.
- **Expand a single term over brackets and collect like terms (pages 2–4):** Use the distributive law, combine like terms, and simplify expressions.
- **Expand the product of two or three expressions (pages 4–6):** Master multiplying binomials and multinomials, using methods such as FOIL and systematic expansion.
- **Factorise linear, quadratic, and simple cubic expressions (pages 6–9):** Reverse the process of expansion to simplify expressions and solve equations.
- **Know and use the laws of indices (pages 9–11):** Use the fundamental rules governing exponents, including products, quotients, powers of powers, and products over products.
- **Simplify and use the rules of surds (pages 12–13):** Work with irrational numbers in radical form and apply their multiplicative properties.
- **Rationalise denominators (pages 13–15):** Eliminate radicals from the denominator using conjugates and the difference of squares.

Before diving into each sub-topic, ensure you are comfortable with basic arithmetic, the distributive property, and combining like terms.

1.1 INDEX LAWS

Aim: To simplify expressions with integer, negative, and fractional powers using a set of core rules.

Fundamental Index Rules

When writing expressions with the same base, the following rules apply:

1. Product of Powers (Same Base):

$$a^m \times a^n = a^{m+n}$$

Example: Simplify $2^3 \times 2^4$. **Solution:** $2^3 \times 2^4 = 2^{3+4} = 2^7 = 128$.

2. Quotient of Powers (Same Base):

$$\left[\frac{a^m}{a^n} = a^{m-n} \quad (\text{provided } a \neq 0)\right]$$

Example: Simplify $\frac{5^6}{5^2}$. **Solution:** $5^6 \div 5^2 = 5^{6-2} = 5^4 = 625$.

3. Power of a Power:

$$[(am)n = a^{m \times n}]$$

Example: Simplify $(3^2)^3$. **Solution:** $(3^2)^3 = 3^{2 \times 3} = 3^6 = 729$.

4. Power of a Product:

$$(ab)^n = a^n \times b^n$$

Example: Simplify $(2 \times 3)^3$. **Solution:** $(2 \times 3)^3 = 2^3 \times 3^3 = 8 \times 27 = 216$.

Visualizing the Process

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Expression: a^m \times a^n
      |
Identify same base: a, with exponents m and n
      |
Apply rule: a^{(m+n)}
      |
Write simplified form: a^{(m+n)}

```

A Quick Reference Table

Property	Formula	Example	Result
Multiplication	$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4$	$2^7 = 128$
Division	$\frac{a^m}{a^n} = a^{m-n}$	$\frac{5^6}{5^2}$	$5^4 = 625$
Power of a Power	$(a^m)^n = a^{m \times n}$	$(3^2)^3$	$3^6 = 729$
Product of a Product	$(ab)^n = a^n \times b^n$	$(2 \times 3)^3$	$2^3 \times 3^3 = 216$

1.2 EXPANDING BRACKETS

Aim: To redistribute terms in an expression and combine like terms to simplify or reassemble the polynomial.

Expanding a Single Term Over Brackets

Use the distributive property:

$$a(b + c) = ab + ac$$

Example: Expand $4(x+3)$. **Solution:** $4(x + 3) = 4x + 12$.

Collecting Like Terms

After expansion, similar variable terms (or constant terms) can be combined: **Example:** Simplify $3x + 4x + 5$. **Solution:** $3x + 4x = 7x$, so the expression becomes $7x + 5$.

Expanding the Product of Two Binomials

When multiplying two binomials, use the FOIL method (First, Outer, Inner, Last):

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

Example: Expand $(x + 2)(x + 3)$:

- **First:** $x \times x = x^2$
- **Outer:** $x \times 3 = 3x$
- **Inner:** $2 \times x = 2x$
- **Last:** $2 \times 3 = 6$ Combine like terms: $x^2 + 3x + 2x + 6 = x^2 + 5x + 6$.

Expanding the Product of Three Expressions

When expanding more than two brackets, work sequentially: **Example:** Expand $(x + 1)(x + 2)(x + 3)$.

1. **Step 1:** Expand the first two binomials.

$$(x + 1)(x + 2) = x^2 + (1 + 2)x + (1 \times 2) = x^2 + 3x + 2$$

2. **Step 2:** Multiply the resulting trinomial by the third binomial:

$$(x^2 + 3x + 2)(x + 3)$$

Distribute each term:

- $x^2 \times x = x^3$
- $x^2 \times 3 = 3x^2$
- $3x \times x = 3x^2$
- $3x \times 3 = 9x$

- $2 \times x = 2x$
- $2 \times 3 = 6$ Now, combine like terms:

$$[x^3 + (3x^2 + 3x^2) + (9x + 2x) + 6 = x^3 + 6x^2 + 11x + 6]$$

Key Tips

- **Systematic Approach:** Write each multiplication step in full before collecting like terms.
- **Check Your Work:** Recombine the expanded terms to ensure no term is omitted.

1.3 FACTORISING

Aim: To reverse the process of expansion. Factorising breaks down an expression into products of simpler expressions that, when multiplied, give the original expression.

Factorising by Taking Out Common Factors

Identify the greatest common factor (GCF) present in all terms: **Example:** Factorise $6x + 9$.

Solution: The GCF is 3, so:

$$6x + 9 = 3(2x + 3)$$

Factorising Quadratic Expressions

For a quadratic in the form:

$$ax^2 + bx + c$$

find two numbers that multiply to $a \times c$ and sum to b .

Example: Factorise $x^2 + 5x + 6$. **Solution:** Find two numbers that multiply to 6 and add to 5: They are 2 and 3. Thus,

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

Factorising the Difference of Two Squares

Any expression of the form:

$$a^2 - b^2$$

factors as:

$$(a + b)(a - b)$$

Example: Factorise $25 - x^2$. **Solution:** Here, $25 = 5^2$ and $x^2 = x^2$, so:

$$25 - x^2 = (5 + x)(5 - x)$$

Factorising Simple Cubic Expressions

Simple cubic expressions sometimes factorise by recognition or by using special cubic formulas such as:

- **Sum of cubes:**

$$[a^3 + b^3 = (a + b)(a^2 - ab + b^2)]$$

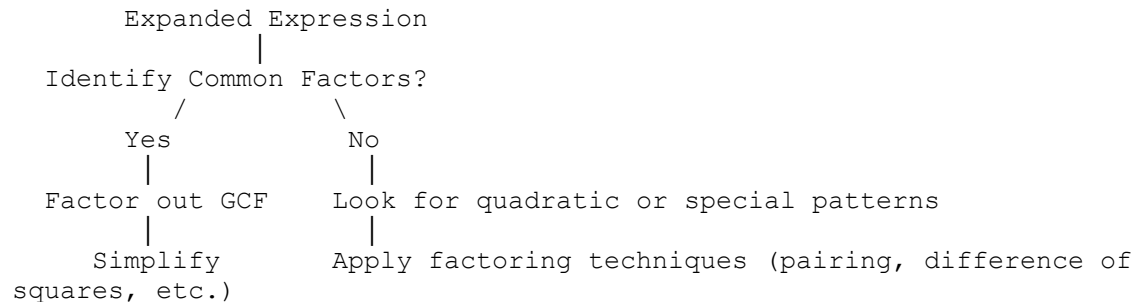
- **Difference of cubes:**

$$[a^3 - b^3 = (a - b)(a^2 + ab + b^2)]$$

Example: Factorise $x^3 - 8$. **Solution:** Since $8 = 2^3$, we have:

$$x^3 - 2^3 = (x - 2)(x^2 + 2x + 4)$$

Visual Flow for Factorising



1.4 NEGATIVE AND FRACTIONAL INDICES

Aim: To extend the laws of indices beyond positive integers to include negative and fractional exponents.

Negative Indices

A negative exponent indicates a reciprocal:

$$a^{-m} = \frac{1}{a^m} \quad (a \neq 0)$$

Example: Simplify 2^{-3} . **Solution:**

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

Zero Exponent

Any non zero number raised to the zero power is 1:

$$a^0 = 1 \quad (a \neq 0)$$

Example: $5^0 = 1$

Fractional Indices

A fractional exponent represents a root. In general:

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

Example 1: Simplify $27^{\frac{2}{3}}$. **Solution:** $\sqrt[3]{27} = 3$ since $3^3 = 27$. Then,

$$27^{\frac{2}{3}} = 3^2 = 9$$

Example 2: Express $16^{\frac{3}{4}}$ in simplest form. **Solution:** $\sqrt[4]{16} = 2$ because $2^4 = 16$, so:

$$16^{\frac{3}{4}} = 2^3 = 8$$

Special Note on Square Roots

For $a^{\frac{1}{2}} = \sqrt{a}$, by convention, we take the principal (positive) square root.

Combining Negative and Fractional Indices

When expressions involve both types, apply the rules sequentially: **Example:** Simplify $8^{-\frac{2}{3}}$.
Solution:

1. First, express the positive fractional exponent: $8^{\frac{2}{3}} = (\sqrt[3]{8})^2 = 2^2 = 4$
2. Then apply the negative: $8^{-\frac{2}{3}} = \frac{1}{4}$

Summary Table for Advanced Indices

Expression	Interpretation	Example: Value Calculation
a^{-m}	Reciprocal: $1/a^m$	$2^{-3} = 1/8$

Expression	Interpretation	Example: Value Calculation
a^0	One, if $a \neq 0$	$5^0 = 1$
$a^{\frac{m}{n}}$	nth root of a^m	$16^{\frac{3}{4}} = (\sqrt[4]{16})^3 = 8$
$a^{\frac{1}{2}}$	Square root of a	$9^{\frac{1}{2}} = \sqrt{9} = 3$

1.5 SURDS

Aim: To understand and simplify irrational numbers expressed in root form, known as surds, and work with them in algebraic expressions.

What Is a Surd?

A surd is an irrational number that cannot be simplified to remove the square root (or other roots). It remains in its radical form.

Rules for Manipulating Surds

1. Multiplication:

$$\sqrt{ab} = \sqrt{a} \times \sqrt{b}$$

Example: Simplify $\sqrt{50}$. **Solution:** $50 = 25 \times 2$, so

$$\sqrt{50} = \sqrt{25 \times 2} = \sqrt{25} \times \sqrt{2} = 5\sqrt{2}$$

2. Division:

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Example: Express $\sqrt{\frac{8}{2}}$ in simplified form. **Solution:** $\sqrt{\frac{8}{2}} = \sqrt{4} = 2$ or alternatively

$$\sqrt{\frac{8}{2}} = \frac{\sqrt{8}}{\sqrt{2}} = \frac{2\sqrt{2}}{\sqrt{2}} = 2.$$

3. Square of a Surd:

$$(a) (\sqrt{a})^2 = a$$

Example: $(\sqrt{7})^2 = 7$.

Expanding Brackets Involving Surds

When surds appear in products or brackets, use the distributive property just as with algebraic terms.

Example: Expand $(2 + \sqrt{3})(3 - \sqrt{3})$. **Solution:**

- Multiply term-by-term:
 - $2 \times 3 = 6$
 - $2 \times (-\sqrt{3}) = -2\sqrt{3}$
 - $3\sqrt{3} \times 3 = 3\sqrt{3}$
 - $\sqrt{3} \times (-\sqrt{3}) = -(\sqrt{3})^2 = -3$
- Then, combine like terms:

$$(2 + \sqrt{3})(3 - \sqrt{3}) = 6 + (-2\sqrt{3} + 3\sqrt{3}) - 3 = 3 + \sqrt{3}$$

Another common example is squaring a binomial with surds:

Example: Expand $(\sqrt{2} + \sqrt{3})^2$. **Solution:**

$$(\sqrt{2} + \sqrt{3})^2 = (\sqrt{2})^2 + 2\sqrt{2\sqrt{3}} + (\sqrt{3})^2 = 2 + 2\sqrt{6} + 3 = 5 + 2\sqrt{6}$$

Why Surds Matter

Surds allow us to work with irrational numbers exactly, rather than approximating decimals, which is crucial for solving equations symbolically and maintaining precision.

1.6 RATIONALISING DENOMINATORS

Aim: To eliminate radicals (surds) from the denominator of a fraction to achieve a more “rational” or simplified form.

Case 1: Denominator is a Single Surd

If an expression has a denominator like $\frac{1}{\sqrt{a}}$, multiply numerator and denominator by $\sqrt{a}$:

Example: Simplify $\frac{1}{\sqrt{2}}$. **Solution:**

$$\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Case 2: Denominator of the Form $a + \sqrt{b}$ or $a - \sqrt{b}$

Multiply by the conjugate of the denominator. The conjugate changes the sign between the two terms.

Example: Simplify $\frac{1}{3+\sqrt{2}}$.

Solution:

1. **Identify the conjugate:** The conjugate of $3 + \sqrt{2}$ is $3 - \sqrt{2}$.
2. **Multiply numerator and denominator:**

$$\frac{1}{3 + \sqrt{2}} \times \frac{3 - \sqrt{2}}{3 - \sqrt{2}} = \frac{3 - \sqrt{2}}{(3)^2 - (\sqrt{2})^2}$$

3. **Apply the difference of squares:**

$$= \frac{3 - \sqrt{2}}{9 - 2} = \frac{3 - \sqrt{2}}{7}$$

Case 3: Denominator with Two Surds, e.g., $\sqrt{a} + \sqrt{b}$

Again, use the conjugate method.

Example: Simplify $\frac{1}{\sqrt{3}+\sqrt{2}}$ **Solution:**

1. **Find the conjugate:** $\sqrt{3} - \sqrt{2}$
2. **Multiply numerator and denominator:**

$$\frac{1}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}} = \frac{\sqrt{3} - \sqrt{2}}{(\sqrt{3})^2 - (\sqrt{2})^2}$$

3. **Simplify denominator using difference of squares:**

$$= \frac{\sqrt{3} - \sqrt{2}}{3 - 2} = \sqrt{3} - \sqrt{2}$$

Visual Summary of Rationalising Denominators

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$$\begin{array}{l} \text{Fraction:} \quad \frac{1}{a + \sqrt{b}} \\ \\ \text{Multiply by:} \quad \frac{(a - \sqrt{b})}{(a - \sqrt{b})} \\ \\ \text{New Denom.:} \quad a^2 - (\sqrt{b})^2 = a^2 - b \end{array}$$

Practical Tips

- Always check if the radical can be simplified before rationalising.
- When multiplying by a conjugate, use the difference of squares to quickly simplify the denominator.

Chapter Review 1

Let's consolidate the key methods you've learned:

1. **Index Laws:**
 - Multiply/divide powers by adding/subtracting exponents.
 - Use the power and product rules to simplify expressions.
2. **Expanding Brackets:**
 - Apply the distributive property.
 - Use FOIL for binomials and sequential multiplication for longer expressions.
3. **Factorising:**
 - Reverse the expansion process by extracting common factors.
 - Recognise special patterns like quadratic trinomials and the difference of squares.
4. **Negative and Fractional Indices:**
 - Interpret negative indices as reciprocals.
 - Understand fractional indices as roots.
5. **Surds:**
 - Simplify radical expressions using multiplication and division rules.
 - Expand and combine surds with algebraic techniques.
6. **Rationalising Denominators:**
 - Remove surds from denominators via multiplication by conjugates and using difference-of-squares identities.

Summary of Key Points

- **Index Laws:** Provide a systematic way to simplify powers, whether multiplying, dividing, or raising a power to a power.
- **Expanding Brackets:** Use distributive property; careful expansion and combination of like terms yield the simplified form.
- **Factorising:** Identify common factors and special patterns to "deconstruct" a polynomial into product form.

- **Negative/Fractional Indices:** Understand that these introduce notions of reciprocals and roots, allowing more general exponent manipulation.
- **Surds:** Maintain precision by leaving irrational numbers in radical form where appropriate and applying key rules.
- **Rationalising:** Ensure final answers are in a conventional form by eliminating radicals from denominators.

2. Quadratics – Detailed Short Notes

Quadratics form one of the most important topics in algebra. Here you will learn how to solve quadratic equations by several methods, understand function notation and behavior, graph quadratic functions, and use the discriminant to determine the nature of the roots. Mastery of these concepts will build a strong foundation not only in algebra but in many applied fields.

Learning Objectives

- **Solve quadratic equations** using factorisation, the quadratic formula, and completing the square (pages 19–24).
- **Read and use $f(x)$ notation** when working with functions (pages 25–27).
- **Sketch the graph and find the turning point** of a quadratic function (pages 27–30).
- **Find and interpret the discriminant** of a quadratic expression (pages 30–32).

Before starting, ensure you're comfortable with algebraic manipulation, rearranging equations, and basic polynomial operations.

2.1 Solving Quadratic Equations

A quadratic equation is generally given in the form:

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

There are several methods to solve such equations.

A. Solving by Factorisation

Method: Express the quadratic as a product of two binomials if possible.

1. **Write in Standard Form:** Ensure the equation is in the form $ax^2 + bx + c = 0$.
2. **Find Two Numbers:** Look for two numbers m and n such that:

$$m \times n = a \times c \quad \text{and} \quad m + n = b$$

3. **Rewrite and Factorise:** Split the middle term using mm and nn, then factor by grouping.

Example 1: Solve $x^2 + 5x + 6 = 0$.

- Here, $a = 1$, $b = 5$, $c = 6$.
- Find numbers that multiply to 6 (i.e. 1×6 or 2×3) and sum to 5: they are 2 and 3.
- Factorisation:

$$x^2 + 5x + 6 = (x + 2)(x + 3) = 0$$

- Set each factor to zero:

$$x + 2 = 0 \quad \Rightarrow \quad x = -2, \quad x + 3 = 0 \quad \Rightarrow \quad x = -3.$$

Example 2: Solve $2x^2 + 7x + 3 = 0$.

- Here, $a = 2$, $b = 7$, $c = 3$.
- Look for two numbers that multiply to $2 \times 3 = 6$ and add to 7: these are 6 and 1.
- Rewrite:

$$2x^2 + 6x + x + 3 = 0.$$

- Factor by grouping:

$$2x(x + 3) + 1(x + 3) = 0.$$

- Factor out the common binomial:

$$(x + 3)(2x + 1) = 0.$$

- Hence, $x = -3$ or $x = -\frac{1}{2}$.

B. Solving Using the Quadratic Formula

When factorisation is not obvious, the quadratic formula is a reliable method. For the quadratic $ax^2 + bx + c = 0$, the solutions are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Example: Solve $2x^2 + 3x - 5 = 0$.

- Here, $a = 2$, $b = 3$, $c = -5$.

- Compute the discriminant:

$$\Delta = b^2 - 4ac = 3^2 - 4(2)(-5) = 9 + 40 = 49.$$

- Apply the formula:

$$x = \frac{-3 \pm \sqrt{49}}{2 \times 2} = \frac{-3 \pm 7}{4}.$$

- This gives two solutions:

$$x = \frac{-3 + 7}{4} = \frac{4}{4} = 1, \quad x = \frac{-3 - 7}{4} = -\frac{10}{4} = -\frac{5}{2}.$$

C. Derivation of the Quadratic Formula (Optional Extension)

To derive the quadratic formula, start with the general quadratic equation:

$$ax^2 + bx + c = 0.$$

Step 1: Divide through by a (assuming $a \neq 0$):

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

Step 2: Rearrange:

$$x^2 + \frac{b}{a}x = -\frac{c}{a}.$$

Step 3: Complete the square by adding $\left(\frac{b}{2a}\right)^2$ to both sides:

$$\left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = -\frac{c}{a} + \frac{b^2}{4a^2}\right]$$

Step 4: Write the left-hand side as a perfect square:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

Step 5: Take the square root of both sides:

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}.$$

Step 6: Solve for x:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

This derivation not only yields the formula but also shows the origin of the discriminant.

2.2 Completing the Square

Completing the square rewrites a quadratic expression into a form that reveals its vertex, making it useful to solve equations and find the maximum or minimum values.

A. Rewriting in the Form $(x + p)^2 + q$

Given a quadratic expression $x^2 + bx + c$ (here, assume the coefficient of x^2 is 1), here's the general procedure:

1. **Identify the coefficient of x:** For $x^2 + bx$, take half of b: $p = \frac{b}{2}$.
2. **Square p:** $p^2 = \left(\frac{b}{2}\right)^2$.
3. **Rewrite the expression:**

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c.$$

Here, $q = c - \left(\frac{b}{2}\right)^2$.

Example: Rewrite $x^2 + 6x + 5$ in completed-square form.

- $b=6$ so $p=\frac{6}{2}=3$ and $p^2=9$.
- Express as:

$$x^2 + 6x + 5 = (x + 3)^2 - 9 + 5 = (x + 3)^2 - 4.$$

- The expression is now in the form $(x + p)^2 + q$ with $p = 3$ and $q = -4$.

B. Solving Quadratic Equations by Completing the Square

Given the equation $x^2 + bx + c = 0$, complete the square to solve.

Example: Solve $x^2 + 6x + 5 = 0$.

1. Write in completed square form:

$$(x + 3)^2 - 4 = 0.$$

2. Rearrange:

$$(x + 3)^2 = 4.$$

3. Take the square root:

$$x + 3 = \pm 2$$

4. Solve for x:

$$x = -3 + 2 = -1 \quad \text{or} \quad x = -3 - 2 = -5.$$

C. Finding the Minimum or Maximum Value of a Quadratic Function

A quadratic function in the form

$$f(x) = a(x + p)^2 + q$$

has:

- A **minimum** if $a > 0$
- A **maximum** if $a < 0$

The vertex $(-p, q)$ gives the minimum or maximum value.

Example: For the function $f(x) = (x + 3)^2 - 4$:

- Since the coefficient of $(x + 3)^2$ is positive (1), the function has a minimum.
- The vertex is $x = -3$ and the minimum value is -4 .

Thus, the smallest value the function can take is -4 , occurring when $x = -3$.

2.3 Functions

Functions are the building blocks of algebra and calculus. Understanding them is essential for analyzing how inputs relate to outputs.

A. Definition of a Function

A **function** f is a rule that assigns to each element x in a set called the **domain** exactly one element $f(x)$ in another set called the **range**. This “input-output” relationship is written as:

$$y = f(x)$$

B. Domain and Range

- **Domain:** The set of all possible inputs x for which the function is defined.
- **Range:** The set of all possible outputs $y = f(x)$.

Example: For $f(x) = \sqrt{x-2}$, the domain is $\{x \mid x \geq 2\}$ and the range is $\{y \mid y \geq 0\}$.

C. $f(x)$ Notation

The notation $f(x)$ denotes the output of the function when x is the input.

Example: If $f(x) = x^2 + 2x + 1$, then:

- $f(3) = 3^2 + 2(3) + 1 = 9 + 6 + 1 = 16$.

D. Composite Functions

A composite function is formed when the output of one function becomes the input of another. If you have two functions f and g , the composite function $f(g(x))$ is defined by:

$$(f \circ g)(x) = f(g(x)).$$

Example: Let $f(x) = 2x$ and $g(x) = x + 3$. Then:

$$f(g(x)) = f(x + 3) = 2(x + 3) = 2x + 6.$$

E. Inverse Functions

An **inverse function** $f^{-1}(x)$ reverses the effect of $f(x)$. If $y = f(x)$, then $f^{-1}(y) = x$.

Procedure to Find $f^{-1}(x)$:

1. Write $y = f(x)$.
2. Swap x and y .
3. Solve for y in terms of x . This new function in y is $f^{-1}(x)$.

Example: Find the inverse of $f(x) = 2x + 3$.

1. Write: $y = 2x + 3$.
2. Swap x and y : $x = 2y + 3$.
3. Solve for y :

$$2y = x - 3 \Rightarrow y = \frac{x - 3}{2}.$$

Thus, $f^{-1}(x) = \frac{x-3}{2}$.

2.4 Quadratic Graphs

Graphing quadratic functions provides a visual insight into the behavior of these functions—specifically, the shape and key features such as the turning point and symmetry.

A. Sketching Quadratic Graphs

Quadratic functions take the form:

$$f(x) = a(x - h)^2 + k. \quad f(x) = ax^2 + bx + c \quad \text{or vertex form} \quad f(x) = a(x - h)^2 + k.$$

Key characteristics:

- **Parabola Shape:** Opens upward if $a > 0$ (minimum point) or downward if $a < 0$ (maximum point).
- **Symmetry:** The graph is symmetric about a vertical line called the **line of symmetry**.

Example: Sketch $f(x) = x^2 - 4x + 3$.

1. **Find the Vertex:**

- Complete the square or use the formula $x = -\frac{b}{2a}$.
- Here, $a=1$, $b=-4$:

$$x = \frac{4}{2} = 2.$$

2. **Calculate $f(2)$:**

$$f(2) = 2^2 - 4(2) + 3 = 4 - 8 + 3 = -1.$$

The vertex is (2,-1).

3. **Line of Symmetry:** The axis is $x=2$.

4. **Intercepts:** Factor: $x^2 - 4x + 3 = (x - 1)(x - 3)$. Roots: $x=1$ and $x=3$; the graph crosses the x -axis at these points.

B. Finding the Turning Point (Vertex) by Completing the Square

Using the vertex form $f(x) = a(x - h)^2 + k$, the turning point is (h,k).

Example: For $f(x) = x^2 + 6x + 5$, we already found:

$$f(x) = (x + 3)^2 - 4.$$

Thus:

- Vertex: (-3,-4).
- The line of symmetry is $x=-3$.

C. Line of Symmetry

As mentioned, for any quadratic function, the vertical line through the vertex is the line of symmetry. For $f(x) = ax^2 + bx + c$, the line is:

$$x = -\frac{b}{2a}.$$

D. Maximum and Minimum Values

- **If $a > 0$:** The parabola opens upward and the vertex is the minimum point.
- **If $a < 0$:** The parabola opens downward and the vertex is the maximum point.

2.5 The Discriminant

The discriminant in the quadratic formula, given by:

$$\Delta = b^2 - 4ac,$$

provides information about the nature and number of solutions.

A. Interpreting the Discriminant

1. $\Delta > 0$: There are two distinct real roots. *Graphically:* The parabola crosses the xx -axis at two points.
2. $\Delta = 0$: There is one repeated real root (also called a double root). *Graphically:* The parabola touches the xx -axis at the vertex.
3. $\Delta < 0$: There are no real roots (two complex conjugate roots). *Graphically:* The parabola does not intersect the xx -axis.

Example 1: Consider $x^2 - 4x + 3 = 0$.

- Compute the discriminant:

$$\Delta = (-4)^2 - 4(1)(3) = 16 - 12 = 4 > 0.$$

- Two real distinct roots.

Example 2: Consider $x^2 - 6x + 9 = 0$.

- Discriminant:

$$\Delta = (-6)^2 - 4(1)(9) = 36 - 36 = 0.$$

- One real repeated root, $x=3$.

Example 3: Consider $x^2 + 2x + 5 = 0$.

- Discriminant:

$$\Delta = 2^2 - 4(1)(5) = 4 - 20 = -16 < 0.$$

- No real roots exist (the roots are complex).

B. Using the Discriminant to Solve Problems

Beyond solving equations, the discriminant can help decide:

- **The nature of the graph:** Knowing whether a parabola intersects the xx-axis.
- **Parametric conditions:** For instance, if a quadratic equation includes a parameter, setting conditions on the discriminant ensures real solutions exist.

Chapter Review 2

Let's summarize the core methods and insights from Quadratics:

1. **Solving by Factorisation:**
 - Identify two numbers whose product equals ac and sum equals b .
 - Factorise by splitting the middle term.
2. **The Quadratic Formula:**
 - Always applicable.
 - Understand the discriminant for insights into the solutions.
3. **Completing the Square:**
 - Rewrite $ax^2 + bx + c$ as $a(x + p)^2 + q$ for easier analysis.
 - Use this technique to solve equations and find the vertex.
4. **Functions and Notation:**
 - Understand domain, range, and the meaning behind $f(x)$.
 - Get comfortable with composite and inverse functions.
5. **Quadratic Graphs:**
 - Sketch parabolas using intercepts, vertex, and symmetry.
 - Determine maximum/minimum values through vertex form.

6. **The Discriminant:**

- Use $\Delta = b^2 - 4ac$ to predict the number and type of solutions.

Summary of Key Points

- **Factorisation, quadratic formula, and completing the square** are three powerful methods to solve quadratic equations.
- **Completing the square** not only solves equations but also unveils the vertex form, offering insights into the graph's turning point.
- **$f(x)$ notation and function properties** are essential for understanding mappings, compositions, and inverses.
- **The graph of a quadratic function** is a parabola characterized by its vertex (turning point), line of symmetry, and intercepts.
- **The discriminant** ($b^2 - 4ac$) quickly informs you whether the quadratic has two, one, or no real solutions.

3. EQUATIONS AND INEQUALITIES

This chapter explores how to solve and interpret both equations and inequalities. You will learn to solve various simultaneous equations (linear–linear, quadratic–quadratic, or one linear and one quadratic) and understand their solutions graphically. In addition, you'll tackle linear and quadratic inequalities and learn to represent their solution sets both numerically (on number lines) and graphically (on coordinate grids).

Learning Objectives

- **Simultaneous Equations:**
 - Solve linear simultaneous equations (pages 37–39)
 - Solve quadratic simultaneous equations (pages 39–40)
 - Solve simultaneous equations where one is linear and one is quadratic (pages 39–40)
 - Interpret algebraic solutions graphically (pages 40–43)
- **Inequalities:**
 - Solve linear inequalities (pages 44–46)
 - Solve quadratic inequalities (pages 46–49)
 - Interpret inequalities graphically (pages 49–51)
 - Represent linear and quadratic inequalities on graphs (pages 49–51)

Before engaging with this chapter, ensure you are well-versed in algebraic manipulation, solving equations, graphing basics, and the concepts behind functions from previous studies.

3.1 LINEAR SIMULTANEOUS EQUATIONS

When two or more linear equations are solved together, the values of the unknowns that satisfy all of them are called the solution of the system. There are two common methods: elimination and substitution.

A. Solving Using Elimination

1. **The Idea:** In the elimination method, you add or subtract equations in order to cancel (eliminate) one of the variables.
2. **Step-by-Step Example:**

Example: Solve

$$\begin{array}{l} 2x + 3y = 12 \quad (1) \\ 4x - 3y = 6 \quad (2) \end{array}$$

Steps:

- Notice that the coefficients of y are $+3$ and -3 . Adding equations (1) and (2):

$$(2x + 3y) + (4x - 3y) = 12 + 6 \Rightarrow 6x = 18.$$

- Solve for x :

$$x = \frac{18}{6} = 3.$$

- Substitute $x = 3$ back into one original equation (say equation (1)):

$$\begin{array}{l} 2(3) + 3y = 12 \Rightarrow 6 + 3y = 12, \\ 3y = 6 \Rightarrow y = 2. \end{array}$$

- **Solution:** $(x, y) = (3, 2)$.

3. Key Tips:

- Ensure like terms are aligned.
- Multiply one or both equations by suitable numbers if coefficients do not cancel immediately.

B. Solving Using Substitution

1. **The Idea:** Isolate one variable in one of the equations and substitute the expression into the other equation.
2. **Step-by-Step Example:**

Example: Solve

$$\begin{aligned}x + 2y &= 7 & (1) \\3x - y &= 5 & (2)\end{aligned}$$

Steps:

- Solve equation (1) for x:

$$x = 7 - 2y.$$

- Substitute $x = 7 - 2y$ into equation (2):

$$\begin{aligned}3(7 - 2y) - y &= 5 & \Rightarrow & 21 - 6y - y = 5, \\21 - 7y &= 5 & \Rightarrow & 7y = 21 - 5 = 16, \\& & & y = \frac{16}{7}.\end{aligned}$$

- Substitute $y = \frac{16}{7}$ back into $x = 7 - 2y$:

$$x = 7 - 2\left(\frac{16}{7}\right) = 7 - \frac{32}{7} = \frac{49 - 32}{7} = \frac{17}{7}.$$

- **Solution:** $(x, y) = \left(\frac{17}{7}, \frac{16}{7}\right)$.

3.2 QUADRATIC SIMULTANEOUS EQUATIONS

When both equations in the system are quadratic, the solution represents the points where the graphs of these curves intersect (which might be none, one, two, or more points).

A. Solving Simultaneous Quadratic Equations

1. **Method: Substitution or Elimination.** First, express one variable in terms of the other (if possible).
2. **Step-by-Step Example:**

Example: Solve the system

$$\begin{aligned}y &= x^2 + 2x + 1, \\y &= -x^2 + 4.\end{aligned}$$

Steps:

- Set the two expressions for y equal:

$$x^2 + 2x + 1 = -x^2 + 4.$$

- Combine like terms:

$$2x^2 + 2x + 1 - 4 = 0 \Rightarrow 2x^2 + 2x - 3 = 0.$$

- Solve the quadratic equation using the quadratic formula:

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(2)(-3)}}{2 \cdot 2} = \frac{-2 \pm \sqrt{4 + 24}}{4} = \frac{-2 \pm \sqrt{28}}{4} = \frac{-2 \pm 2\sqrt{7}}{4} = \frac{-1 \pm \sqrt{7}}{2}.$$

- Substitute each value of x back into one of the original equations (e.g., $y = -x^2 + 4$) to find y .

For $x = \frac{-1+\sqrt{7}}{2}$:

$$y = -\left(\frac{-1+\sqrt{7}}{2}\right)^2 + 4.$$

Compute the square and simplify (details omitted for brevity—but work step-by-step to get the exact coordinate).

- **Interpretation:** The solution(s) represent the intersection points of the two parabolic curves.
3. **Graphical Perspective:** When plotted, each curve (parabola) will intersect at the computed point(s).

3.3 SIMULTANEOUS EQUATIONS ON GRAPHS

Graphing simultaneous equations gives a visual representation of their solutions.

A. Interpreting Solutions as Points of Intersection

1. **Concept:** Each equation is graphed on the coordinate plane. The point(s) where the graphs meet correspond to the solution(s) of the system.
2. **Example:** Consider the system:

$$\begin{aligned} y &= 2x + 1, \\ y &= -x + 4 \end{aligned}$$

- Graph $y = 2x + 1$ (a line with slope 2) and $y = -x + 4$ (a line with slope -1).
- The intersection point can be found algebraically or by sketching.
- Solving:

$$2x + 1 = -x + 4 \Rightarrow 3x = 3 \Rightarrow x = 1,$$

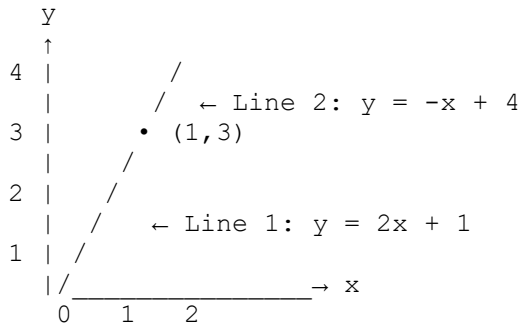
$$y = 2(1) + 1 = 3.$$

- The intersecting point is (1, 3).

B. Solving By Sketching Graphs

1. **Technique:** Draw rough but accurate graphs for each equation.
2. **Tips:**
 - Clearly mark the intercepts.
 - Use the line of symmetry (for quadratic graphs) if needed.
 - The sketch should visually confirm the algebraic solution(s).

Visual Sketch (Conceptual):



3.4 LINEAR INEQUALITIES

A linear inequality relates a linear expression with another using inequality signs ($<$, $\leq$, $>$, $\geq$). The goal is to find all values of the variable that satisfy the inequality.

A. Solving Linear Inequalities

1. **Principle:** The same rules as equations apply—except when multiplying or dividing by a negative number, the inequality sign must be flipped.
2. **Example:** Solve:

$$-2x + 5 > 1.$$

Steps:

- Subtract 5 from both sides:

$$-2x > 1 - 5, \quad -2x > -4.$$

- Divide by -2 and flip the sign:

$$x < \frac{-4}{-2} \Rightarrow x < 2.$$

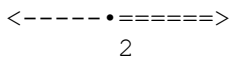
B. Representing Solutions on a Number Line

1. How to Represent:

- Draw a horizontal line.
- Mark the boundary value (e.g., $x=2$).
- Use an open circle if the inequality is strict ($<$ or $>$) or a closed circle for inclusive inequalities ($\leq$ or $\geq$).
- Shade the portion corresponding to the solution.

2. Illustration:

Number Line:



Here, an open circle at 2 with shading to the left represents $x < 2$.

C. Solving Inequalities with More Than One Part

1. Example: Solve:

$$1 < 3x + 2 \leq 11.$$

Steps:

- Subtract 2 from all parts:

$$1 - 2 < 3x \leq 11 - 2 \Rightarrow -1 < 3x \leq 9.$$

- Divide through by 3:

$$-\frac{1}{3} < x \leq 3.$$

- **Solution:** x is greater than $-1/3$ and less than or equal to 3.

D. Contextual Problems Involving Linear Inequalities

1. **Example:** A company has to produce at least 100 units and can produce up to 500 units per day. Let x be the number of units produced. Write and solve the inequality.

$$100 \leq x \leq 500.$$

2. **Interpretation:** The solution is the set of all x between 100 and 500 (inclusive). Graph this on a number line to visualize the feasible production range.

3.5 QUADRATIC INEQUALITIES

A quadratic inequality involves a quadratic expression set in relation to zero (or another expression) by an inequality sign.

A. Solving Quadratic Inequalities by Factorisation

1. **Steps:**
 - Express the quadratic inequality in factored form.
 - Find the roots (values where the expression equals zero).
 - Divide the number line into intervals based on these roots.
 - Test a value from each interval in the inequality to determine the sign.
2. **Example:** Solve:

$$x^2 - 5x + 6 > 0.$$

Solution:

- Factor:

$$x^2 - 5x + 6 = (x - 2)(x - 3).$$

- Find roots: $x = 2$ and $x = 3$.
- Test intervals:
 - For $x < 2$ (e.g., $x = 0$): $(0 - 2)(0 - 3) = (+6) > 0$.
 - For $2 < x < 3$ (e.g., $x = 2.5$): $(0.5)(-0.5) = -0.25 < 0$.
 - For $x > 3$ (e.g., $x = 4$): $(4 - 2)(4 - 3) = (2)(1) = 2 > 0$.
- **Solution:** $x < 2$ or $x > 3$.

B. Using a Sketch of the Quadratic Graph

1. **Method:** Graph the quadratic function $y = x^2 - 5x + 6$.
 - Find the vertex and x-intercepts.
 - Identify the regions where $y > 0$ (i.e., the graph is above the x-axis).
2. **Verification:** The graph will lie above the x-axis for $x < 2$ and $x > 3$, confirming our solution.

3.6 INEQUALITIES ON GRAPHS

Graphing inequalities gives you a visual representation of all the solutions.

A. Graphical Interpretation of Linear Inequalities

1. **Example (Linear):** Graph $x + 2 > 3$.

Steps:

- Solve: $x > 1$.
 - On the number line, mark 1 with an open circle and shade to the right.
2. **Graph on a Coordinate Plane:** If the inequality involves two variables (e.g., $y > 2x + 1$), draw the boundary line $y = 2x + 1$ (dashed if not included) and shade the appropriate half-plane (above the line if y is greater).

B. Graphical Interpretation of Quadratic Inequalities

1. **Example (Quadratic):** Graph $y \leq x^2 - 5x + 6$.

Steps:

- Graph the parabola $y = x^2 - 5x + 6$.
- Shade the region **below** the parabola (since the inequality is “less than or equal to”).
- The boundary curve (the parabola) is drawn as a solid line if equality is included ($\leq$ or $\geq$).

3.7 REGIONS

When working with systems of inequalities, the solution is often a region on the coordinate plane defined by the overlapping areas of each inequality.

A. Representing Regions Defined by Inequalities on a Coordinate Grid

1. **Method:**
 - Graph each inequality on the same coordinate plane.
 - The solution to the system is the intersection (overlap) of the shaded regions.
2. **Example:** Consider the system:

$$\begin{aligned}y &\geq 2x + 1, \\ y &\leq -x + 5\end{aligned}$$

Steps:

- Graph $y = 2x + 1$ (solid line, shading above it) and $y = -x + 5$ (solid line, shading below it).
- The overlapping area (the region satisfying both conditions) is the solution set.

Chapter Review 3

Let's consolidate the key ideas:

1. **Linear Simultaneous Equations:**
 - Master both elimination and substitution methods.
 - Translate contextual problems into systems and solve them with the appropriate method.
2. **Quadratic Simultaneous Equations:**
 - When both equations are quadratic, set them equal (or substitute) to find intersections.
 - Recognize that the solution's nature (number of points) is linked to the curves' intersection.
3. **Graphical Methods for Simultaneous Equations:**
 - Use graphs to visually interpret the points where equations intersect.
4. **Linear Inequalities:**
 - Solve using algebraic manipulations (remember to flip the sign when dividing by negatives).
 - Represent the solution on a number line and apply these techniques to real-life problems.
5. **Quadratic Inequalities:**
 - Factorise the quadratic, find its roots, and test intervals.
 - Graph the quadratic function to determine where it is greater or less than a given value.
6. **Graphing Inequalities and Regions:**
 - Understand that the solution region for a system of inequalities is the overlap of the individual solution sets.
 - Practice shading regions on both number lines (for one-variable problems) and coordinate grids (for two-variable problems).

Summary of Key Points

- **Simultaneous Equations:**
 - **Linear systems:** Use elimination or substitution; contextual word problems translate to these systems.
 - **Quadratic systems:** Set equations equal, solve algebraically, and interpret the solutions as intersections.
 - **Graphing:** Always check your algebraic solutions with a rough sketch for verification.
- **Inequalities:**
 - Solve linear and quadratic inequalities systematically.
 - Always flip the inequality sign when multiplying/dividing by a negative.
 - Represent solutions both numerically (number lines) and graphically (coordinate grids).
 - Understand that graphing systems of inequalities helps visualize the feasible region.
- **Regions on Graphs:** Representing solution sets on a coordinate grid builds intuition about constraints—

Learning Objectives

- **Sketch cubic graphs:** Understand the shape of cubic curves, find their roots and intercepts, and sketch them from factorised forms. *(Pages 58–61)*
- **Sketch reciprocal graphs:** Learn to draw the graphs of $y = \frac{1}{x}$ and $y = \frac{a}{x^2}$, identify their asymptotes, and understand their behavior. *(Pages 62–63)*
- **Use intersection points of graphs to solve equations:** Interpret the algebraic solutions as intersections on the coordinate plane. *(Pages 63–66)*
- **Translate graphs:** Move graphs vertically and horizontally by applying translations. *(Pages 67–71)*
- **Stretch graphs:** Apply vertical and horizontal stretches (or compressions) to alter the graph's shape. *(Pages 71–74)*
- **Transform graphs of unfamiliar functions:** Combine multiple operations to transform graphs step by step. *(Pages 75–77)*

Before starting, ensure you are comfortable with basic function plotting, solving simple equations, and the graphing of linear and quadratic functions.

4.1 CUBIC GRAPHS

A. General Shape of Cubic Graphs

- **Standard Form:** A general cubic equation takes the form:

$$y = ax^3 + bx^2 + cx + d, \quad \text{with } a \neq 0.$$

- **Behaviour at Extremes:**
 - When $a > 0$: As $x \rightarrow +\infty, y \rightarrow +\infty$; as $x \rightarrow -\infty, y \rightarrow -\infty$.
 - When $a < 0$: The behaviour is reversed.
- **Characteristic S-Shape:** Most cubic graphs display an S-shaped curve. Unlike quadratics, cubic functions can have one or two turning points (points where the direction of the curve changes).

B. Finding Roots and Intercepts

- **X-intercepts (Roots):** These are the values of x where $y = 0$. When the cubic is written in factorised form, for example:

$$y = a(x - r_1)(x - r_2)(x - r_3),$$

then r_1, r_2 , and r_3 are the roots.

- **Y-intercept:** Found by setting $x = 0$; i.e., $y = d$ in the form $y = ax^3 + bx^2 + cx + d$.

C. Sketching Cubic Graphs from Their Factorised Form

- **Example 1: Sketch**

$$y = (x - 1)(x + 2)(x - 3).$$

Steps:

1. **Identify the roots:** $x = 1$, $x = -2$, and $x = 3$. Mark these on the x-axis.
2. **Determine the y-intercept:** Set $x = 0$:

$$y = (-1)(2)(-3) = 6.$$

3. **Assess the end behavior:** Since the coefficient of x^3 (when expanded) is positive, as $x \rightarrow \infty, y \rightarrow \infty$; as $x \rightarrow -\infty, y \rightarrow -\infty$.
 4. **Sketch:**
 - Draw a smooth curve passing through $(-2, 0)$, $(1, 0)$, and $(3, 0)$.
 - Ensure the curve passes through $(0, 6)$ and respects the end behavior.
- **Key Points:**
 - Mark the intercepts.
 - Sketch the curve smoothly with the expected turning points.
 - Verify the end behavior with the sign of the leading coefficient.

4.2 RECIPROCAL GRAPHS

A. Sketching the Graph of $y = \frac{1}{x}$

- **Shape and Behaviour:**
 - The graph is a hyperbola divided into two branches.
 - **Quadrant Locations:** One branch is in the first quadrant (when $x > 0$ and $y > 0$); the other is in the third quadrant (when $x < 0$ and $y < 0$).
- **Key Features:**
 - **Vertical Asymptote:** $x = 0$ (the graph never touches the y-axis).
 - **Horizontal Asymptote:** $y = 0$ (as $x \rightarrow \pm\infty$).
- **Example Calculation:**
 - For $x = 1, y = 1$.
 - For $x = -2, y = -\frac{1}{2}$.

B. Sketching the Graph of $y = \frac{a}{x^2}$

- **Shape and Behaviour:**
 - Regardless of whether x is positive or negative, x^2 is always positive. Thus, if $a > 0$, then $y > 0$ for all $x \neq 0$.

- **Symmetry:** Graph is symmetric about the y-axis.
- **Key Features:**
 - **Vertical Asymptote:** $x = 0$.
 - **Horizontal Asymptote:** $y = 0$.
- **Example Calculation with $a = 2$:**
 - For $x = 1, y = \frac{2}{1} = 2$.
 - For $x = -2, y = \frac{2}{4} = 0.5$.

C. Asymptotes of Reciprocal Graphs

- **Vertical Asymptote:**
 - Occur when the denominator equals zero; for both $y = \frac{1}{x}$ and $y = \frac{a}{x^2}, x = 0$.
- **Horizontal Asymptote:**
 - Occurs when x tends to plus or minus infinity; here, $y = 0$ in both cases.
- **Graphical Tip:**
 - Draw dashed lines for the asymptotes to emphasize the boundaries the graph will approach but never cross.

4.3 POINTS OF INTERSECTION

A. Solving Equations by Finding Intersection Points

- **Concept:** If you have two functions, $y = f(x)$ and $y = g(x)$, their points of intersection satisfy $f(x) = g(x)$. Graphically, these are the points where the curves cross.
- **Example:** Solve for x by finding the intersection of:

$$f(x) = x^2 \quad \text{and} \quad g(x) = 4x - 3.$$

Steps:

1. Set the functions equal:

$$x^2 = 4x - 3 \quad \Rightarrow \quad x^2 - 4x + 3 = 0.$$

2. Factorise:

$$(x - 1)(x - 3) = 0.$$

3. Find x -values:

$$x = 1 \quad \text{or} \quad x = 3.$$

4. Determine corresponding y -values:

For $x = 1$:

$$y = 1^2 = 1 \Rightarrow (1, 1).$$

For $x = 3$:

$$y = 3^2 = 9 \Rightarrow (3, 9).$$

- **Graphical Interpretation:** The solution set $\{(1, 1), (3, 9)\}$ represents points where the parabola $y = x^2$ meets the line $y = 4x - 3$.

4.4 TRANSLATING GRAPHS

A. Translation Parallel to the Y-Axis: $y = f(x) + a$

- **Concept:** Adding a constant a to function $f(x)$ shifts the graph vertically.
 - If $a > 0$, the graph shifts **up**.
 - If $a < 0$, the graph shifts **down**.
- **Example:** For $f(x) = x^2$:
 - $f(x) + 3 = x^2 + 3$ shifts the parabola upward by 3 units.
 - $f(x) - 4 = x^2 - 4$ shifts the parabola downward by 4 units.

B. Translation Parallel to the X-Axis: $y = f(x + a)$

- **Concept:** Here, $f(x + a)$ shifts the graph horizontally.
 - If $a > 0$, the graph shifts **left** by a units.
 - If $a < 0$, the graph shifts **right**.
- **Example:** For $f(x) = x^2$:
 - $f(x + 2) = (x + 2)^2$ shifts the parabola to the left by 2 units.
 - $f(x - 3) = (x - 3)^2$ shifts the parabola to the right by 3 units.

C. Combining Translations

- **Example:** Consider $y = (x - 2)^2 + 3$.
 - The $(x - 2)^2$ indicates a horizontal translation to the **right** by 2 units.
 - The $+3$ indicates a vertical translation **up** by 3 units.
 - The vertex of the standard parabola $y = x^2$ moves from $(0, 0)$ to $(2, 3)$.

4.5 STRETCHING GRAPHS

A. Stretch Parallel to the Y-Axis: $y = a \cdot f(x)$

- **Concept:** Multiplying $f(x)$ by a constant a affects the y -values.
 - If $|a| > 1$: The graph stretches vertically.

- If $0 < |a| < 1$: The graph compresses vertically.
- **Example:** Take $f(x) = x^2$:
 - $y = 2x^2$ makes the parabola “narrower” by stretching vertically.
 - $y = \frac{1}{2}x^2$ makes it “wider” by compressing vertically.

B. Stretch Parallel to the X-Axis: $y = f(ax)$

- **Concept:** Multiplying the input x by a affects the horizontal spread.
 - If $|a| > 1$: The graph compresses horizontally.
 - If $0 < |a| < 1$: The graph stretches horizontally.
- **Example:** With $f(x) = x^2$:
 - $y = (2x)^2 = 4x^2$ compresses the graph horizontally.
 - $y = \left(\frac{1}{2}x\right)^2 = \frac{1}{4}x^2$ stretches it horizontally.

4.6 TRANSFORMING FUNCTIONS

Applying Multiple Transformations

Transforming a graph often requires two or more operations—translations, reflections, and stretches/compressions. The order in which you apply these operations can be important.

- **General Strategy:**
 1. **Horizontal Shifts/Stretch:** Begin with any changes inside the function argument (shifts $f(x \pm a)$, horizontal stretches $f(bx)$).
 2. **Vertical Stretch/Compression and Reflection:** Next, adjust the output—multiplication by a factor, adding/subtracting constants.
 3. **Vertical Shifts:** Finally, apply y-axis translations through additions/subtractions outside the function.
- **Example:** Transform the function $y = x^2$ into

$$y = -3(x + 2)^2 + 4.$$

Step-by-Step Analysis:

- **Horizontal Translation:** $(x+2)$ shifts the graph 2 units to the **left**.
- **Vertical Stretch and Reflection:** Multiplying by -3:
 - The factor of 3 stretches the graph vertically (it becomes “narrower”).
 - The negative sign reflects the graph about the x-axis.
- **Vertical Translation:** Adding +4 shifts the entire graph **up** by 4 units.
- **Result:** The vertex of $y = x^2$, originally at $(0,0)$, moves to $(-2, 4)$. The parabola opens downward (because of the negative factor) and is steeper than the original.

Chapter Review 4

Let's summarize the key ideas and methods covered in this chapter:

1. **Cubic Graphs:**
 - Understand the general S-shape of cubics.
 - Find roots by factorising and determine intercepts.
 - Sketch graphs from their factorised form while respecting turning points and end behavior.
2. **Reciprocal Graphs:**
 - Sketch $y = \frac{1}{x}$ with its hyperbolic branches and clearly mark asymptotes.
 - Sketch $y = \frac{a}{x^2}$, noting symmetry and asymptotic behavior.
3. **Points of Intersection:**
 - Solve equations by equating functions to find intersection points.
 - Use both algebraic methods and graph sketches to confirm solutions.
4. **Translations:**
 - Shift graphs vertically ($y = f(x) + a$) and horizontally ($y = f(x + a)$).
 - Combine shifts to move graphs without changing their shape.
5. **Stretching Graphs:**
 - Apply vertical stretches/compressions via $y = af(x)$.
 - Adjust the horizontal scale with $y = f(ax)$.
6. **Transforming Functions:**
 - Apply multiple transformations in sequence.
 - Understand the effect of each transformation on the graph's geometry.

Summary of Key Points

- **Cubic Graphs:**
 - Recognize their basic "S" shape and the role of factorisation in finding intercepts.
 - Use end behavior (dictated by the leading coefficient) to guide your sketches.
- **Reciprocal Graphs:**
 - Identify and draw asymptotes (vertical at $x=0$ and horizontal at $y=0$).
 - Understand the distinct behaviors of $y = \frac{1}{x}$ versus $y = \frac{a}{x^2}$.
- **Intersection Points:**
 - Solve equations graphically by finding where two graphs meet.
 - Verify algebraic solutions with clear sketches.
- **Graph Translations:**
 - Vertical shifts ($y = f(x) + a$) move the graph up/down.
 - Horizontal shifts ($y = f(x + a)$) move the graph left/right.
 - Combined translations reposition the function's key points (e.g., the vertex of a parabola).
- **Graph Stretching:**
 - Vertical stretching ($y = af(x)$) modifies the height.
 - Horizontal stretching ($y = f(ax)$) alters the width.

- Adjusting these factors can change the “steepness” or “compression” of the graph.
- **Multiple Transformations:**
 - Transform challenging functions by breaking down each operation.
 - Consistently apply the operations in the correct order to achieve the desired graph.

5. STRAIGHT LINE GRAPHS

Straight lines form the backbone of analytic geometry. In this chapter, you will learn how to calculate the gradient, write and manipulate the equations of lines in various forms, and use these skills to solve problems involving intersections, distances, and areas on coordinate grids. Mastery of these concepts is essential for both pure mathematics and applications in fields ranging from physics to economics.

Learning Objectives

- **Calculate the gradient:** Learn to compute the gradient (slope) of a line joining two points. *(Pages 86–87)*
- **Link equation with gradient and intercept:** Understand how the equation of a line directly corresponds to its slope and where it crosses the y-axis. *(Pages 87–89)*
- **Find the equation of a line:** Determine a line’s equation given either (i) its gradient and one point, or (ii) two distinct points. *(Pages 89–92)*
- **Parallel and perpendicular lines:** Know and apply the rules governing parallel (equal gradient) and perpendicular lines (product of gradients equals -1). *(Pages 93–96)*
- **Intersection points:** Determine the intersection of two lines by solving simultaneous equations. *(Pages 97–99)*
- **Length and area on grids:** Solve geometric problems on coordinate grids, such as calculating the distance between points and the area of a triangle defined by coordinates. *(Pages 96–99)*

Before proceeding, ensure you are comfortable with basic algebra, coordinate plotting, and solving simple equations.

5.1 $y = mx + c$

The slope-intercept form is the cornerstone for understanding straight line graphs.

A. Gradient of a Line Joining Two Points

Definition: For two points (x_1, y_1) and (x_2, y_2) on a line, the gradient m is calculated as:

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad \text{provided that } x_2 \neq x_1.$$

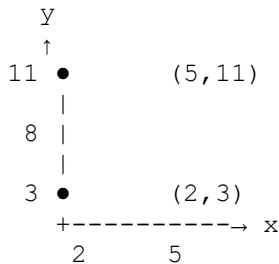
Understanding as Rate of Change: The gradient represents the rate at which y changes relative to x . A positive m means the line is increasing; a negative m indicates the line is decreasing. The magnitude of m tells you the steepness.

Example 1: Find the gradient of the line through $(2, 3)$ and $(5, 11)$:

$$m = \frac{11 - 3}{5 - 2} = \frac{8}{3} \approx 2.67.$$

This means for every 3 units increase in x , y increases by 8 units.

Diagram (Conceptual):



The "rise" between the points is 8 and the "run" is 3.

5.2 EQUATIONS OF STRAIGHT LINES

Different forms of a line's equation are used depending on what information is available.

A. Slope-Intercept Form: $y = mx + c$

- m = gradient (slope)
- c = y -intercept (the point $(0, c)$ where the line crosses the y -axis)

Example 2: For the line $y = 2x + 3$:

- Gradient $m = 2$ (the line rises 2 units for every 1 unit increase in x).
- y -intercept $c = 3$, so the line crosses $(0, 3)$.

B. Point-Slope Form: $y - y_1 = m(x - x_1)$

This is especially useful when you know a point (x_1, y_1) and the gradient m .

Derivation: Starting with the gradient formula,

$$m = \frac{y - y_1}{x - x_1} \Rightarrow y - y_1 = m(x - x_1).$$

Example 3: Find the equation of a line passing through (3, 4) with a gradient of -2:

$$y - 4 = -2(x - 3). y - 4 = -2(x - 3).$$

Simplify:

$$y = -2x + 6 + 4 \Rightarrow y = -2x + 10.$$

C. Finding the Equation Given Two Points

1. **Calculate the gradient:** Use $m = \frac{y_2 - y_1}{x_2 - x_1}$.
2. **Apply the point-slope form:** Substitute one point (x_1, y_1) with the computed mm.

Example 4: Find the equation through (1, 2) and (4, 8).

- Compute $m = \frac{8-2}{4-1} = \frac{6}{3} = 2$.
- Substitute into point-slope form (using (1,2)):

$$y - 2 = 2(x - 1) \Rightarrow y = 2x - 2 + 2 = 2x.$$

D. Standard Form: $ax + by + c = 0$

This form is useful for solving systems of equations.

Example 5: Rewrite $y = 2x + 3$ in standard form:

$$y - 2x - 3 = 0 \quad \text{or equivalently} \quad 2x - y + 3 = 0.$$

5.3 PARALLEL AND PERPENDICULAR LINES

A. Parallel Lines

Rule: Lines are parallel if they have the same gradient.

$$m_1 = m_2.$$

Example 6: If L_1 is given by $y = 3x + 1$, then any line parallel to L_1 must have the form $y = 3x + c$, where c is a constant (the y-intercept may differ).

B. Perpendicular Lines

Rule: Lines are perpendicular if the product of their gradients equals -1 :

$$m_1 \times m_2 = -1.$$

Example 7: Find a line perpendicular to $y = 2x + 5$. Here, $m_1 = 2$, so the perpendicular line must have gradient:

$$m_2 = -\frac{1}{2}.$$

The general form is $y = -\frac{1}{2}x + c$.

C. Finding Equations for Parallel and Perpendicular Lines

Parallel Example: Find the equation of a line parallel to $y = -3x + 2$ that passes through (4, 1).

- Since the new line is parallel, its gradient is -3 .
- Use point-slope form:

$$y - 1 = -3(x - 4) \Rightarrow y = -3x + 12 + 1 \Rightarrow y = -3x + 13.$$

Perpendicular Example: Find the equation of a line perpendicular to $y = \frac{1}{4}x - 3$ that passes through (8, 2).

- The gradient of the given line is $\frac{1}{4}$, so the perpendicular gradient is:

$$m = -4 \quad \left(\text{since } \frac{1}{4} \times -4 = -1 \right).$$

- Use point-slope form:

$$y - 2 = -4(x - 8) \Rightarrow y = -4x + 32 + 2 \Rightarrow y = -4x + 34.$$

5.4 LENGTH AND AREA ON COORDINATE GRIDS

A. Distance Between Two Points

The distance formula—derived from the Pythagorean theorem—calculates the distance d between points (x_1, y_1) and (x_2, y_2) :

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Example 8: Find the distance between (1, 2) and (4, 6):

$$d = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5.$$

B. Finding the Area of a Triangle Using Coordinates

If a triangle's vertices are given by (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) , the area Δ of the triangle can be calculated as:

$$\Delta = \frac{1}{2} \text{Big} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \text{Big}.$$

Example 9: Find the area of the triangle with vertices at $(1,2)$, $(4,6)$, and $(7,2)$.

Calculate:

$$\begin{aligned} \Delta &= \frac{1}{2} \text{Big} |1(6-2) + 4(2-2) + 7(2-6)| \text{Big} = \frac{1}{2} \text{Big} |4 + 0 - 28| \text{Big} = \frac{1}{2} \times 24 \\ &= 12. \end{aligned}$$

C. Finding the Point of Intersection of Two Lines

To determine where two lines intersect, set their equations equal and solve the resulting system of equations.

Example 10: Find the intersection of $L_1: y = 2x + 3$ and $L_2: y = -x + 9$.

1. Set the equations equal:

$$2x + 3 = -x + 9.$$

2. Solve for x :

$$3x = 6 \Rightarrow x = 2.$$

3. Substitute $x = 2$ back into L_1 :

$$y = 2(2) + 3 = 7.$$

Thus, the lines intersect at $(2, 7)$.

Chapter Review 5

Let's consolidate the key points:

1. **Gradient Calculation:**

- The gradient, $m = \frac{y_2 - y_1}{x_2 - x_1}$, measures the steepness (rate of change) of the line.

2. Equations of Lines:

- **Slope-Intercept Form:** $y = mx + c$ shows the slope and y-intercept.
- **Point-Slope Form:** $y - y_1 = m(x - x_1)$ is useful when one point and the gradient are known.
- **Standard Form:** $ax + by + c = 0$ is often used in systems of equations.

3. Parallel and Perpendicular Lines:

- Parallel lines share the same gradient.
- Perpendicular lines have gradients that multiply to -1 ($m_1 \times m_2 = -1$).

4. Distance and Area Calculations:

- Use the distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, to find lengths.
- Use the coordinate formula for the area of a triangle to solve area problems on grids.

5. Intersection Points:

- Solve simultaneous equations to find where two lines intersect.

Summary of Key Points

- **Gradient as Rate of Change:** The gradient provides both direction and steepness.
- **Multiple Equation Forms:**
 - $y = mx + c$ (slope-intercept) shows clear physical meaning with gradient and intercept.
 - $y - y_1 = m(x - x_1)$ (point-slope) and $ax + by + c = 0$ (standard form) offer flexibility in handling different problems.
- **Parallelism & Perpendicularity:**
 - Parallel lines have $m_1 = m_2$.
 - Perpendicular lines satisfy $m_1 \times m_2 = -1$.
- **Coordinate Geometry Applications:**
 - The distance formula and the area formula are practical tools for solving geometrical problems on grids.
 - The intersection of lines is determined algebraically by solving simultaneous equations.

6. TRIGONOMETRIC RATIOS

This chapter unifies the study of triangles with the powerful tools of trigonometry. You will learn how to apply the cosine and sine rules to determine missing sides or angles, calculate triangle areas using trigonometric formulas, and solve practical problems involving triangles. In addition, you will explore the graphs of the sine, cosine, and tangent functions and understand how simple transformations—translations, stretches, and reflections—affect their shape.

Learning Objectives

- **Use the cosine rule** to find a missing side or angle in a triangle. (*Pages 105–110*)
- **Use the sine rule** to find a missing side or angle in a triangle, including handling the ambiguous case. (*Pages 110–116*)
- **Find the area of a triangle** using an appropriate formula ($\frac{1}{2} ab \sin C$). (*Pages 116–118*)
- **Solve problems involving triangles** by choosing the best trigonometric approach. (*Pages 118–122*)
- **Sketch the graphs** of the sine, cosine, and tangent functions and understand their key features. (*Pages 123–125*)
- **Sketch simple transformations** of these trigonometric graphs (translations, stretches, reflections). (*Pages 125–129*)

Before proceeding, ensure you have a strong foundation in basic trigonometry (defining sine, cosine, and tangent in right-angled triangles) and familiarity with solving basic algebraic equations.

6.1 THE COSINE RULE

The cosine rule is an essential tool for solving any triangle (not just right-angled ones). It relates the lengths of the sides of a triangle to the cosine of one of its angles.

A. Finding a Missing Side

The cosine rule for finding a side is given by:

$$a^2 = b^2 + c^2 - 2bc \cos A,$$

where a , b , and c are the sides of the triangle, and A is the angle opposite side a .

Example 1: Given a triangle with sides $b = 8$ cm and $c = 10$ cm and an included angle $A = 60^\circ$, find the side a .

Steps:

1. Substitute into the cosine rule:

$$a^2 = 8^2 + 10^2 - 2 \times 8 \times 10 \cos 60^\circ.$$

2. Calculate:

$$8^2 = 64, \quad 10^2 = 100, \quad \cos(60^\circ) = 0.5.$$

3. Then:

$$a^2 = 64 + 100 - 2 \times 8 \times 10 \times 0.5 = 164 - 80 = 84.$$

4. So:

$$a = \sqrt{84} \approx 9.17 \text{ cm.}$$

B. Finding a Missing Angle

When all three sides are known, you can find an angle using the cosine rule rearranged as:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}.$$

Example 2: Given a triangle with sides $a = 7$ cm, $b = 9$ cm, and $c = 12$ cm, find angle A .

Steps:

1. Substitute into the rearranged cosine rule:

$$\cos A = \frac{9^2 + 12^2 - 7^2}{2 \times 9 \times 12}.$$

2. Calculate:

$$9^2 = 81, \quad 12^2 = 144, \quad 7^2 = 49.$$

3. Then:

$$\cos A = \frac{81 + 144 - 49}{216} = \frac{176}{216} \approx 0.8148.$$

4. Finally,

$$A = \cos^{-1}(0.8148) \approx 35.0^\circ.$$

C. Solving Problems Using the Cosine Rule

The cosine rule is particularly useful when you know two sides and the included angle (SAS case) or when you know all three sides (SSS case). – **Pointers:**

- Always check that the given angle is the one *between* the two known sides.
- Verify the reasonableness of computed sides/angles with known geometric bounds.

6.2 THE SINE RULE

The sine rule provides an alternative method for solving triangles, particularly useful when you know two angles and one side (AAS or ASA cases) or two sides and a non-included angle (SSA case).

A. The Sine Rule Statement

The sine rule states:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Here, a , b , and c are the sides of the triangle, and A , B , and C are the angles opposite these sides, respectively.

B. Finding a Missing Side or Angle

Example 3: In a triangle, suppose $A = 45^\circ$, $B = 60^\circ$, and side $a = 10 \text{ cm}$. Find side b .

Steps:

1. First, find angle C (using $A + B + C = 180^\circ$):

$$C = 180^\circ - 45^\circ - 60^\circ = 75^\circ.$$

2. Use the sine rule:

$$\frac{10}{\sin 45^\circ} = \frac{b}{\sin 60^\circ}.$$

3. Substitute known values:

$$\sin 45^\circ \approx 0.7071, \quad \sin 60^\circ \approx 0.8660.$$

4. So:

$$\frac{10}{0.7071} \approx 14.14 = \frac{b}{0.8660},$$
$$b \approx 14.14 \times 0.8660 \approx 12.25 \text{ cm. } b \approx 14.14 \times 0.8660 \approx 12.25 \text{ cm.}$$

C. The Ambiguous Case

The “ambiguous case” occurs with the SSA condition, where you are given two sides and a non-included angle.

Example 4: Suppose you know that $a = 8 \text{ cm}$, $b = 10 \text{ cm}$, and $A = 30^\circ$.

- First, use the sine rule:

$$\frac{8}{\sin 30^\circ} = \frac{10}{\sin B}.$$

- Since $\sin 30^\circ = 0.5$:

$$\frac{8}{0.5} = 16 = \frac{10}{\sin B} \Rightarrow \sin B = \frac{10}{16} = 0.625.$$

- B can be:

$$B \approx \sin^{-1}(0.625) \approx 38.7^\circ,$$

or the supplement:

$$B' = 180^\circ - 38.7^\circ = 141.3^\circ.$$

- Check with the triangle sum:
 - For $B \approx 38.7^\circ$, then

$$C = 180^\circ - 30^\circ - 38.7^\circ \approx 111.3^\circ,$$

which is valid.

- For $B' \approx 141.3^\circ$, then

$$C = 180^\circ - 30^\circ - 141.3^\circ = 8.7^\circ,$$

which is also possible depending on the given context.

- **Note:** The ambiguous case tells us that with SSA data, two distinct triangles (or sometimes only one) might satisfy the conditions.

6.3 AREAS OF TRIANGLES

One of the most efficient formulas for finding the area of a triangle when you know two sides and the included angle is based on the sine function.

A. The Area Formula

The area (Δ) of a triangle is given by:

$$\text{Area} = \frac{1}{2}ab \sin C,$$

where a and b are two sides and C is the included angle.

Example 5: Find the area of a triangle with sides $a = 7$ cm, $b = 9$ cm, and an included angle $C = 45^\circ$.

Steps:

1. Substitute in the formula:

$$\text{Area} = \frac{1}{2} \times 7 \times 9 \times \sin 45^\circ.$$

2. Recall that $\sin 45^\circ \approx 0.7071$.
3. Compute:

$$\text{Area} = \frac{1}{2} \times 63 \times 0.7071 \approx 31.5 \times 0.7071 \approx 22.3 \text{ cm}^2.$$

B. Using the Formula in Problem Solving

Besides the straightforward use, this formula is particularly useful when neither the base nor the height is readily known, as it directly relates the sides with the included angle.

6.4 SOLVING TRIANGLE PROBLEMS

When solving triangle problems, it is crucial to choose the appropriate trigonometric rule based on the given information.

A. Choosing the Appropriate Rule

- **When Sides and Included Angle (SAS) are Known:** Use the cosine rule to find the missing side, then apply the sine rule for the remaining angles if needed.
- **When Two Angles and One Side (AAS or ASA) are Known:** Apply the sine rule directly.
- **When Two Sides and a Non-Included Angle (SSA) are Known:** Use the sine rule and be cautious of the ambiguous case.

Example 6: A triangle has sides of 12 cm and 15 cm with an included angle of 70° .

1. Use the cosine rule to find the third side.
2. Then use the sine rule to find one of the unknown angles.
3. Finally, compute the remaining angle using the triangle sum property.

B. Solving Various Triangle Problems

Practice problems may include:

- Determining side lengths, angle measures, and area.
- Resolving ambiguous cases.
- Applying these methods in real-world contexts such as navigation or surveying.

6.5 GRAPHS OF SINE, COSINE, AND TANGENT

Understanding the graphs of trigonometric functions is fundamental for visualizing periodic behavior.

A. The Graph of $y = \sin x$

- **Key Features:**
 - **Amplitude:** 1 (the maximum distance from the midline).
 - **Period:** 2π (the function repeats every 2π).
 - **Midline:** $y = 0$.
 - **Zeros:** At $x = 0, \pi, 2\pi, \dots$
 - **Maximum:** $y = 1$ at $x = \frac{\pi}{2}$; **Minimum:** $y = -1$ at $x = \frac{3\pi}{2}$.

Sketch Tip: Plot key points over one period and draw a smooth, continuous curve.

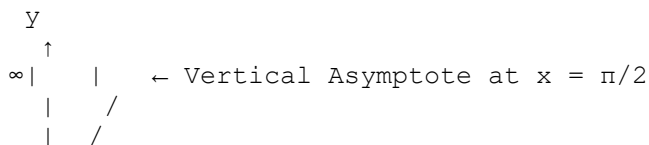
B. The Graph of $y = \cos x$

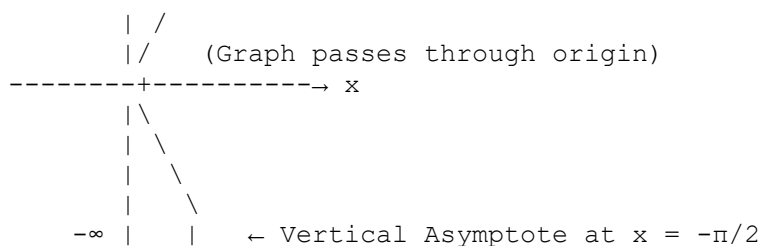
- **Key Features:**
 - **Amplitude:** 1.
 - **Period:** 2π .
 - **Zeros:** At $x = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$
 - **Maximum:** $y = 1$ at $x = 0, 2\pi, \dots$

C. The Graph of $y = \tan x$

- **Key Features:**
 - **Period:** π .
 - **Asymptotes:** Vertical asymptotes occur at $x = \frac{\pi}{2} + k$ (where the cosine is zero).
 - **Behavior:** The function increases steeply between the asymptotes, crossing the origin (for $y = \tan x$).

Diagram (Conceptual for $\tan x$):





6.6 TRANSFORMING TRIGONOMETRIC GRAPHS

Transformations allow you to modify the basic trigonometric graphs into new forms. These include translations, stretches (compressions), and reflections.

A. Translations

- **Vertical Translation:** $y = \sin x + k$ shifts the graph up (if $k > 0$) or down (if $k < 0$).
Example: $y = \cos x - 2$ shifts the cosine graph down by 2 units.
- **Horizontal Translation (Phase Shift):** $y = \sin(x - h)$ shifts the graph to the right by h units; if h is negative, the shift is to the left. *Example:* $y = \cos\left(x + \frac{\pi}{4}\right)$ shifts the cosine graph to the left by $\frac{\pi}{4}$.

B. Stretches

- **Vertical Stretch/Compression:** $y = A \sin x$ changes the amplitude. If $|A| > 1$, the graph stretches vertically; if $0 < |A| < 1$, it compresses. *Example:* $y = 3 \sin x$ triples the amplitude.
- **Horizontal Stretch/Compression:** $y = \sin(Bx)$ changes the period of the function. The period becomes $\frac{2\pi}{|B|}$. *Example:* $y = \sin(2x)$ has a period of π .

C. Reflections

- **Reflection Across the X-Axis:** $y = -\sin x$ or $y = -\cos x$ flips the graph vertically.
- **Reflection Across the Y-Axis (for cosine, which is even):** Typically not altering $\cos x$, but note that sine, an odd function, satisfies $y = \sin(-x) = -\sin x$.

D. Combining Transformations

Often, you must apply more than one transformation simultaneously. Follow a consistent order:

1. Apply horizontal shifts and stretches inside the function argument.
2. Apply vertical stretches/compressions and then vertical translations.
3. Reflect if needed.

Example 7: Transform $y = \sin x$ to

$$y = -2 \sin \left(2 \left(x - \frac{\pi}{4} \right) \right) + 3.$$

Interpretation:

- **Inside the sine function:**
 - $\left(x - \frac{\pi}{4} \right)$ shifts the graph right by $\frac{\pi}{4}$.
 - The multiplier 2 horizontally compresses the period from 2π to π .
- **Outside the sine function:**
 - The factor -2 vertically stretches the graph by 2 and reflects it over the x-axis.
 - The +3 shifts the entire graph up 3 units.
- **Result:** The new graph has a modified amplitude, period, phase shift, and vertical translation, all of which can be checked by plotting key points.

Chapter Review 6

Let's consolidate the key ideas covered in this chapter:

1. **Cosine Rule:**
 - Used to find a missing side ($a^2 = b^2 + c^2 - 2bc \cos A$) or a missing angle ($\cos A = \frac{b^2 + c^2 - a^2}{2bc}$).
 - Applies to SAS and SSS cases.
2. **Sine Rule:**
 - Expresses the relation: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$.
 - Very useful when working with ASA, AAS, or SSA (ambiguous case) problems.
3. **Area of Triangle:**
 - The formula $\text{Area} = \frac{1}{2}ab \sin C$ provides an efficient method when two sides and the included angle are known.
4. **Solving Triangle Problems:**
 - Determine which rule (sine or cosine) is appropriate based on the given information.
5. **Graphs of Trigonometric Functions:**
 - Understand the standard shape, amplitude, period, zeros, and asymptotes for $\sin x$, $\cos x$, and $\tan x$.
6. **Transformations:**
 - Translate, stretch, and reflect trigonometric graphs to model shifts in amplitude, period, and phase.

Summary of Key Points

- **Cosine Rule:** Essential for non-right triangles; applicable in both SSS and SAS cases.
- **Sine Rule:** Provides a straightforward relation between sides and angles; always check for the ambiguous case.

- **Area Calculation:** $\frac{1}{2}ab \sin C$ is a direct method when two sides and the included angle are known.
- **Graphing:**
 - $y = \sin x$ and $y = \cos x$ share a period of 2π , while $y = \tan x$ has a period of π .
 - Recognize key features such as amplitude, period, zeros, and asymptotes.
- **Transformations of Trigonometric Graphs:**
 - Vertical/horizontal translations change the location.
 - Stretches/compressions modify the amplitude or period.
 - Reflections invert the graph across the x-axis (or y-axis for odd functions).

7. RADIANS

Trigonometry and the geometry of circles become far more elegant when angles are measured in radians rather than degrees. In this chapter, you will learn the definition of a radian, how to interconvert between degrees and radians, and how to use radian measure to solve problems relating to arc lengths and areas of sectors or segments of circles.

Learning Objectives

- **Convert between degrees and radians:** Understand the definition of a radian, perform conversions using the key equality π radians $= 180^\circ$, and know the exact radian values for common angles (30° , 45° , 60° , 90°). (*Pages 134–135*)
- **Find an arc length using radians:** Employ the formula $l = r\theta$ (with θ measured in radians) to determine the length of an arc on a circle. (*Pages 135–139*)
- **Find areas of sectors and segments using radians:** Use the formulas $A = \frac{1}{2}r^2\theta$ for sectors and $A = \frac{1}{2}r^2(\theta - \sin \theta)$ for segments to solve area problems. (*Pages 139–145*)

Before you proceed, ensure that you are comfortable with the basic concepts of circles, degrees, and simple algebraic manipulations.

7.1 RADIAN MEASURE

A. Definition of a Radian

- **What Is a Radian?** One radian is defined as the angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle.

In other words, if a circle has radius r , and an arc of that circle has length r , then the central angle that intercepts that arc is exactly 1 radian.

B. Conversion between Degrees and Radians

- **Key Conversion Factor:**

$$\pi \text{ radians} = 180^\circ$$

- **Conversion Formulas:**

- **Degrees to Radians:** Multiply by $\frac{\pi}{180}$.

$$\text{Radians} = \text{Degrees} \times \frac{\pi}{180}.$$

- **Radians to Degrees:** Multiply by $\frac{180}{\pi}$.

$$\text{Degrees} = \text{Radians} \times \frac{180}{\pi}.$$

C. Exact Radian Values for Common Angles

- **30°:**

$$30^\circ = 30 \times \frac{\pi}{180} = \frac{\pi}{6} \text{ radians.}$$

- **45°:**

$$45^\circ = 45 \times \frac{\pi}{180} = \frac{\pi}{4} \text{ radians.}$$

- **60°:**

$$60^\circ = 60 \times \frac{\pi}{180} = \frac{\pi}{3} \text{ radians.}$$

- **90°:**

$$90^\circ = 90 \times \frac{\pi}{180} = \frac{\pi}{2} \text{ radians.}$$

Example: Convert 75° to radians.

$$75^\circ \times \frac{\pi}{180} = \frac{75\pi}{180} = \frac{5\pi}{12} \text{ radians.}$$

7.2 ARC LENGTH

The arc length of a fraction of a circle is directly related to the measure of the central angle in radians.

A. Formula for Arc Length

For a circle of radius r and a central angle θ (expressed in radians), the arc length l is given by:

$$l = r\theta.$$

This formula is derived from the proportionality between the angle θ and the full circle's circumference $2\pi r$.

B. Solving Problems Involving Arc Length

Example 1: Find the length of the arc for a circle with $r = 5$ cm and a central angle $\theta = \frac{\pi}{3}$.

Solution:

$$l = 5 \times \frac{\pi}{3} = \frac{5\pi}{3} \text{ cm.}$$

Example 2: A circle has a radius of 10 cm. What is the arc length corresponding to a central angle of 120° ?

Step 1: Convert 120° to radians:

$$120^\circ = 120 \times \frac{\pi}{180} = \frac{2\pi}{3} \text{ radians.}$$

Step 2: Compute arc length:

$$l = 10 \times \frac{2\pi}{3} = \frac{20\pi}{3} \text{ cm.}$$

7.3 AREAS OF SECTORS AND SEGMENTS

Knowing the area of a section of a circle is vital in many applications. We will study both sectors (the “pie-slice” portion of a circle) and segments (the area between a chord and the corresponding arc).

A. Area of a Sector

When the central angle θ is given in radians, the area A of a sector of a circle with radius r is:

$$A_{\text{sector}} = \frac{1}{2}r^2\theta.$$

Example 3: Find the area of a sector for a circle of radius 6 cm with a central angle of 60° .

Step 1: Convert 60° to radians:

$$60^\circ = \frac{\pi}{3} \text{ radians.}$$

Step 2: Apply the formula:

$$A = \frac{1}{2} \times 6^2 \times \frac{\pi}{3} = \frac{1}{2} \times 36 \times \frac{\pi}{3} = 18 \times \frac{\pi}{3} = 6\pi \text{ cm}^2.$$

B. Area of a Segment

A segment of a circle is the region between a chord and the arc that it subtends. Its area can be found by subtracting the area of the triangle (formed by the two radii and the chord) from the area of the corresponding sector.

An alternate direct formula for the area of a segment with central angle θ (in radians) is:

$$A_{\text{segment}} = \frac{1}{2} r^2 (\theta - \sin \theta).$$

Example 4: Find the area of a segment in a circle with radius 4 cm and a central angle of 90° .

Step 1: Convert 90° to radians:

$$90^\circ = \frac{\pi}{2} \text{ radians.}$$

Step 2: Substitute into the area formula:

$$A_{\text{segment}} = \frac{1}{2} \times 4^2 \left(\frac{\pi}{2} - \sin \frac{\pi}{2} \right).$$

Step 3: Calculate:

- $4^2 = 16$,
- $\sin \frac{\pi}{2} = 1$.

Thus:

$$A_{\text{segment}} = \frac{1}{2} \times 16 \left(\frac{\pi}{2} - 1 \right) = 8 \left(\frac{\pi}{2} - 1 \right) = 4\pi - 8 \text{ cm}^2.$$

C. Solving Problems Involving Areas of Sectors and Segments

Practice problems may involve:

- Finding the area of a sector and then determining the area of a segment when a triangular portion is removed.
- Combining these formulas in composite figures where parts of circles are involved with other geometric shapes.

Chapter Review 7

Let's consolidate what we have covered:

1. Radian Measure:

- A radian is the angle for which the arc length equals the radius.
- Conversion between degrees and radians uses the relation π radians = 180° .
- Common angles:
 - $30^\circ = \frac{\pi}{6}$,
 - $45^\circ = \frac{\pi}{4}$,
 - $60^\circ = \frac{\pi}{3}$,
 - $90^\circ = \frac{\pi}{2}$.

2. Arc Length:

- The formula $l = r\theta$ is used (with θ in radians) to find the length of an arc.
- The arc length is directly proportional both to the radius and to the measure of the central angle.

3. Areas of Sectors and Segments:

- **Sector:** $A_{\text{sector}} = \frac{1}{2}r^2$ gives the area of a “pie-slice” section of a circle.
- **Segment:** $A_{\text{segment}} = \frac{1}{2}r^2(\theta - \sin \theta)$ finds the area between a chord and its intercepted arc.
- These formulas are particularly useful in problems where the circle is not “whole” but only part of the geometry.

Summary of Key Points

- **Radians vs. Degrees:** Use the conversion $\text{Radians} = \text{Degrees} \times \frac{\pi}{180}$ and $\text{Degrees} = \text{Radians} \times \frac{180}{\pi}$. Knowing common angle measures in radians (e.g., $30^\circ = \frac{\pi}{6}$) is essential.
- **Arc Length:** The relationship $l = r\theta$ simplifies analysis by putting angle measurements directly in terms of the circular arc's length.
- **Area Formulas:**
 - Sector area: $A = \frac{1}{2}r^2$.

- Segment area: $A = \frac{1}{2}r^2(\theta - \sin \theta)$. These are critical for solving problems related to portions of circles.
- Problem-Solving:** With clear formulas and conversion steps, you can confidently tackle varied problems involving circular portions—from design questions to real-world applications in architecture, engineering, and beyond.

8. DIFFERENTIATION

Differentiation is the process of finding the instantaneous rate of change of a function—its derivative. This concept not only tells us how a function is changing at any point but also helps us understand geometric properties of curves (like the slope of a tangent line) and analyze the behavior of functions. In this chapter we focus on three main tasks:

- Finding the derivative, $f'(x)$ (or $\frac{dy}{dx}$) of a simple function.
- Using the derivative to solve problems involving gradients, tangents, and normals.
- Finding the second derivative, $f''(x)$ (or $\frac{d^2y}{dx^2}$), which informs us about the curvature and concavity of a function.

Before you begin, make sure you're comfortable with the concept of limits and basic algebraic manipulation.

Learning Objectives

- Find the derivative, $f'(x)$ or $\frac{dy}{dx}$, of a simple function.** (Pages 154–163)
- Use the derivative to solve problems involving gradients, tangents, and normals.** (Pages 163–165)
- Find the second derivative, $f''(x)$ or $\frac{d^2y}{dx^2}$, of a simple function.** (Pages 165–166)

8.1 GRADIENTS OF CURVES

A. Gradient of a Chord

- Definition:** A chord is a straight line joining two points on a curve. Its gradient (slope) between points (x_1, y_1) and (x_2, y_2) is given by:

$$m_{\text{chord}} = \frac{y_2 - y_1}{x_2 - x_1}.$$

- **Example:** For the function $f(x) = x^2$, find the gradient of the chord joining the points where $x = 2$ and $x = 5$. Compute:

$$y_1 = 2^2 = 4, \quad y_2 = 5^2 = 25,$$

so,

$$m_{\text{chord}} = \frac{25 - 4}{5 - 2} = \frac{21}{3} = 7.$$

B. Gradient of a Tangent to a Curve

- **Concept:** The tangent to a curve at a given point is the line that “just touches” the curve there. Its gradient is defined as the limit of the gradient of a chord as the two points draw infinitely close.

8.2 DIFFERENTIATING x^n

A. The Power Rule

- **Rule Statement:** For any real number n ,

$$\frac{d}{dx}(x^n) = n \cdot x^{n-1}.$$

- **Example 3:** Find the derivative of $f(x) = x^5$:

$$f'(x) = 5x^{5-1} = 5x^4.$$

- **Example 4:** Differentiate $f(x) = x^{-3}$:

$$f'(x) = -3x^{-4} = -\frac{3}{x^4}.$$

This rule follows directly from the limit definition and is the cornerstone of differentiating polynomials.

8.3 DIFFERENTIATING QUADRATICS

A. Differentiating Quadratic Functions

Quadratic functions have the form:

$$f(x) = ax^2 + bx + c.$$

Apply the power rule to each term:

- **Example 5:** For $f(x) = 3x^2 + 4x + 5$:
 - Derivative of $3x^2$: $3 \times 2x = 6x$.
 - Derivative of $4x$: 4.
 - The constant 5 differentiates to 0.

Therefore,

$$f'(x) = 6x + 4.$$

8.4 DIFFERENTIATING FUNCTIONS WITH TWO OR MORE TERMS

A. Term-by-Term Differentiation

Differentiation is a linear operation. For a function:

$$f(x) = g(x) + h(x),$$

the derivative is:

$$f'(x) = g'(x) + h'(x).$$

- **Example 6:** Differentiate $f(x) = 2x^3 - 5x^2 + 7x - 4$.
 - Derivative of $2x^3$: $2 \times 3x^2 = 6x^2$.
 - Derivative of $-5x^2$: $-5 \times 2x = -10x$.
 - Derivative of $7x$: 7.
 - The constant -4 yields 0.

So,

$$f'(x) = 6x^2 - 10x + 7.$$

B. Rearrangement if Required

Often, functions are already expressed in the form ax^n so that each term is directly differentiable using the power rule.

8.5 GRADIENTS, TANGENTS, AND NORMALS

A. Finding the Gradient at a Point Using the Derivative

- **Method:** If $f(x)$ is differentiable, the gradient of the curve at $x = x_1$ is $f'(x_1)$.

- **Example 7:** Given $f(x) = x^2 + 3x$, first differentiate:

$$f'(x) = 2x + 3.$$

At $x = 2$:

$$f'(2) = 2(2) + 3 = 7.$$

Hence, the gradient at $x = 2$ is 7.

B. Equation of the Tangent Line

- **Formula:** The equation of the tangent to the curve at $(x_1, f(x_1))$ is given by:

$$y - f(x_1) = f'(x_1)(x - x_1).$$

- **Example 8:** For $f(x) = x^2 + 3x$ at $x = 2$:
 - Compute the point: $f(2) = 2^2 + 3(2) = 4 + 6 = 10$, so the point is $(2, 10)$.
 - The gradient is 7 (as above).
 - Hence, the tangent line is:

$$y - 10 = 7(x - 2) \quad \text{or} \quad y = 7x - 4.$$

C. Equation of the Normal Line

- **Definition:** The normal line is perpendicular to the tangent.
- **Rule:** If the tangent has gradient m , then the normal has gradient $-\frac{1}{m}$.
- **Example 9:** For the same function with tangent gradient 7 at $(2, 10)$, the normal's gradient is:

$$m_{\text{normal}} = -\frac{1}{7}.$$

Its equation is:

$$y - 10 = -\frac{1}{7}(x - 2).$$

8.6 SECOND ORDER DERIVATIVES

A. Finding the Second Derivative

The second derivative $f''(x)$ is the derivative of the first derivative $f'(x)$. It indicates the curvature or concavity of the original function.

- **Notation:** $f''(x)$ or $\frac{d^2y}{dx^2}$.
- **Example 10:** For $f(x) = x^2 + 3x$, recall:

$$f'(x) = 2x + 3.$$

Differentiate $f'(x)$ to obtain:

$$f''(x) = \frac{d}{dx}(2x + 3) = 2.$$

Since $f''(x) = 2$ is positive (and constant), the graph of $f(x)$ is concave upward everywhere.

CHAPTER REVIEW 8

Let's consolidate the key points from this chapter:

1. **Gradients of Curves:**
 - A chord's gradient is computed as the difference quotient between two points.
 - The gradient of the tangent (or instantaneous rate of change) is the limit of the chord gradients, leading to the definition of the derivative.
2. **Finding the Derivative from First Principles:**
 - The derivative is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

- We illustrated this with $f(x) = x^2$ and $f(x) = x^3$.

3. **The Power Rule:**
 - For any function $f(x) = x^n$,

$$f'(x) = nx^{n-1}.$$

- This rule simplifies differentiating polynomials.

4. **Differentiating Quadratic and Multi-Term Functions:**
 - Differentiate each term separately using the linearity of differentiation.
5. **Using the Derivative for Geometric Applications:**
 - The derivative at a point gives the gradient.
 - The tangent line at $(x_1, f(x_1))$ is given by $y - f(x_1) = f'(x_1)(x - x_1)$.
 - The normal line is perpendicular to the tangent, with slope $-\frac{1}{f'(x_1)}$.

6. **Second Order Derivatives:**
 - They are found by differentiating the first derivative and provide insight into the concavity of the function.

SUMMARY OF KEY POINTS

- **Definition of Derivative:** The derivative is the limit of the difference quotient as the increment tends to zero.
- **Power Rule:** Differentiating x^n is accomplished using the rule nx^{n-1} .
- **Linearity:** Differentiating a sum or difference of functions means differentiating each term separately.
- **Application to Geometry:** Derivatives provide the gradient, enabling determination of tangents and normals.
- **Second Derivative:** Provides information on curvature and concavity; for example, a positive second derivative indicates the function is concave up.

9. INTEGRATION

Integration is essentially the reverse process of differentiation. It allows you to “accumulate” the values of a function and is used to find areas under curves, recover original functions from their derivatives, and solve many practical problems. In this chapter, we focus on three main tasks:

- **Finding $\int x^n dx$ for x^n** (Pages 171–173)
- **Integrating polynomials** (Pages 172–175)
- **Finding $f(x)$ given $f'(x)$ and a point on the curve** (Pages 176–178)

Before starting, ensure you are comfortable with the basics of differentiation and the concept of an antiderivative.

9.1 INTEGRATING x^n

A. Integration as the Reverse of Differentiation

- **Key Idea:** If differentiation tells you how fast a function is changing at each point, integration “accumulates” that change to recover the original function (up to a constant). In other words, integration is the inverse process of differentiation.

B. The Rule for Integrating x^n

- **Fundamental Rule:** For any real number $n \neq -1$,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c,$$

where c is the **constant of integration**.

- **Why $n \neq -1$?** When $n = -1$, the integral becomes $\int \frac{1}{x} dx$, which is defined separately as:

$$\int \frac{1}{x} dx = \ln|x| + c.$$

C. The Constant of Integration, c

- **Explanation:** Because differentiation of any constant is 0, when you integrate a function, the constant that was “lost” during differentiation cannot be recovered. Thus, every antiderivative is determined only up to an arbitrary constant c.

D. Examples

1. **Example 1:** Find $\int x^2 dx$. **Solution:** Using the power rule with $n = 2$:

$$\int x^2 dx = \frac{x^{2+1}}{2+1} + c = \frac{x^3}{3} + c.$$

2. **Example 2:** Compute $\int x^{-3} dx$. **Solution:** Here $n = -3$:

$$\int x^{-3} dx = \frac{x^{-3+1}}{-3+1} + c = \frac{x^{-2}}{-2} + c = -\frac{1}{2x^2} + c.$$

9.2 INDEFINITE INTEGRALS OF POLYNOMIALS

A. Indefinite Integrals Explained

- **Definition:** An indefinite integral represents the family of all antiderivatives of a function. For a polynomial $P(x)$,

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

the indefinite integral is found by integrating each term separately.

B. Integration Term by Term

- **Rule:**

$$\begin{aligned} & \int [a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0] dx \\ &= a_n \int x^n dx + a_{n-1} \int x^{n-1} dx + \cdots + a_1 \int x dx + a_0 \int dx. \end{aligned}$$

C. Example

Example 4: Integrate the polynomial:

$$\int (3x^2 + 2x - 5)dx.$$

Step-by-Step:

1. **Separate the integral:**

$$\int 3x^2 dx + \int 2x dx - \int 5 dx.$$

2. **Integrate each term:**

- $\int 3x^2 dx = 3 \cdot \frac{x^3}{3} = x^3.$
- $\int 2x dx = 2 \cdot \frac{x^2}{2} = x^2.$
- $\int 5 dx = 5x.$

3. **Combine and add constant cc:**

$$\int (3x^2 + 2x - 5)dx = x^3 + x^2 - 5x + c.$$

9.3 FINDING FUNCTIONS GIVEN THEIR DERIVATIVE

A. Concept: Antidifferentiation to Recover the Original Function

- **Problem Statement:** Given $f'(x)$ and a point (x_0, y_0) on $f(x)$, we wish to recover $f(x)$. This process is sometimes called solving an **initial value problem**.

B. Method

1. **Integrate $f'(x)$ to Obtain the General Form of $f(x)$.**
2. **Use the Given Point (x_0, y_0) to Determine the Constant c .**

C. Example

Example 5: Suppose $f'(x) = 3x^2 + 2$ and you know that $f(1) = 7$. Find $f(x)$.

Solution:

1. **Integrate $f'(x)$:**

$$f(x) = \int (3x^2 + 2)dx = 3 \int x^2 dx + 2 \int dx.$$

- $\int x^2 dx = \frac{x^3}{3}$, so $\int x^2 dx = 3 \left(\frac{x^3}{3} \right) = x^3$.
- $\int dx = x$, so $\int dx = 2x$.

Thus,

$$f(x) = x^3 + 2x + c.$$

2. **Determine c using $f(1) = 7$:**

$$f(1) = 1^3 + 2(1) + c = 1 + 2 + c = 3 + c.$$

Since $f(1) = 7$, we have:

$$3 + c = 7 \quad \Rightarrow \quad c = 4.$$

3. **Write the final function:**

$$f(x) = x^3 + 2x + 4.$$

CHAPTER REVIEW 9

Let's consolidate the key ideas of this chapter:

1. **Integration as the Reverse Process of Differentiation:**
 - Finding the antiderivative simply “undoes” differentiation.
2. **Power Rule for Integration:**
 - For any exponent $n \neq -1$,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c.$$

3. **Indefinite Integrals of Polynomials:**
 - Use linearity and the power rule to integrate each term individually.
 - Always include the constant of integration, c.
4. **Recovering a Function from Its Derivative:**
 - Integrate the derivative and use any provided initial condition to solve for c.

SUMMARY OF KEY POINTS

- **Definition and Method:** Integration is the reverse process of differentiation. It is expressed as finding an antiderivative such that, when differentiated, it recovers the original function.
- **The Power Rule for Integration:** $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ for $n \neq -1$.

- **Indefinite Integrals:** When integrating a polynomial, each term is integrated separately and then combined, always adding the constant c .
- **Recovery of a Function:** Given $f'(x)$ and a point on the curve, integrate to find $f(x)$ and use the point to determine the constant of integration.